

PROOF SEARCH IN NATURAL DEDUCTION

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under the supervision of
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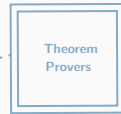
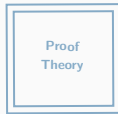
SPLoGIC, University of Campinas

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Proof Search in Natural Deduction: Main Idea

In our research project, we intend to close a gap coming from two ends: the theory of natural deduction and a well-established theorem prover based on natural deduction - *Isabelle*.

Modeling of
Human Reasoning



Human Readable Outputs
and Proof Formats

Proof Theory: Modeling of Human Reasoning

It all started with **Hilbert**, who first developed his proof theory to be used in meta-mathematical applications. However, he seemed to have a broader goal in mind...

Modeling of
Human Reasoning

Proof
Theory



The fundamental idea
of my proof theory is
none other than to
describe the activity of our
understanding, to make a
protocol of the rules according
to which our thinking activity
actually proceeds.

Theorem
Provers

Human Readable Outputs
and Proof Formats

Proof Theory: Modeling of Human Reasoning

In 1935, **Gentzen** set up the system of Natural Deduction, "a formalism that reflects as accurately as possible the actual logical reasoning involved in mathematical proofs".

Modeling of
Human Reasoning

Proof
Theory



Natural Deduction

General and Intuitive Logic

$$\begin{aligned} & x = \frac{2 \cdot 2}{2 \cdot 2} = \frac{2 \cdot 2}{2 \cdot 2} \cdot x \\ & \frac{2}{2 \cdot 2} \cdot x = \frac{2}{2 \cdot 2} \cdot x \\ & \frac{2}{2 \cdot 2} \cdot x = \frac{2}{2 \cdot 2} \cdot x \\ & \frac{2}{2 \cdot 2} \cdot x = \frac{2}{2 \cdot 2} \cdot x \\ & \frac{2}{2 \cdot 2} \cdot x = \frac{2}{2 \cdot 2} \cdot x \\ & \frac{2}{2 \cdot 2} \cdot x = \frac{2}{2 \cdot 2} \cdot x \\ & \frac{2}{2 \cdot 2} \cdot x = \frac{2}{2 \cdot 2} \cdot x \end{aligned}$$



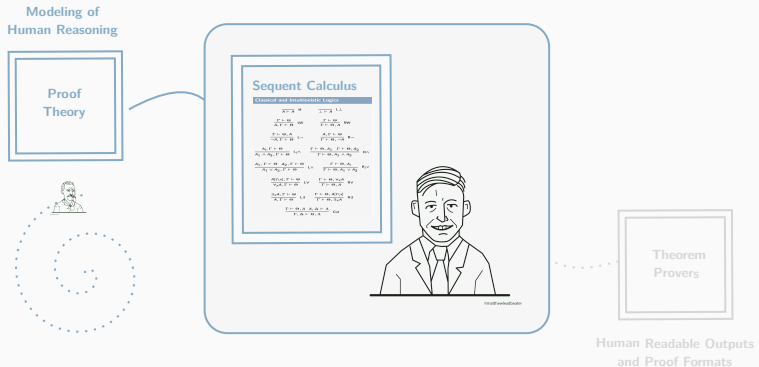
gentzen

Theorem
Provers

Human Readable Outputs
and Proof Formats

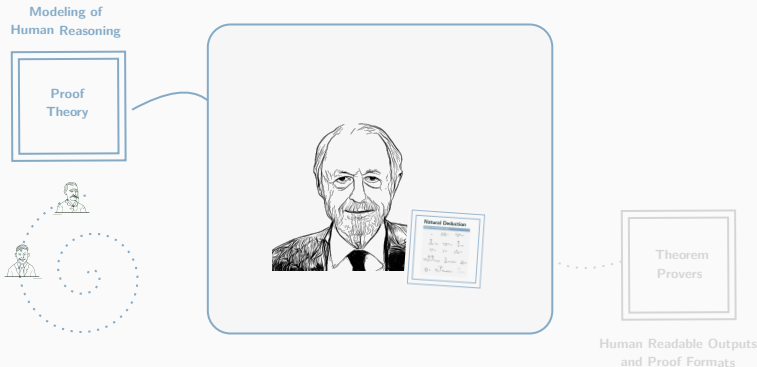
Proof Theory: Modeling of Human Reasoning

Nevertheless, **Gentzen** immediately introduced another formalism, the Sequent Calculus, which, contrary to Natural Deduction, offers a theory of reasoning.



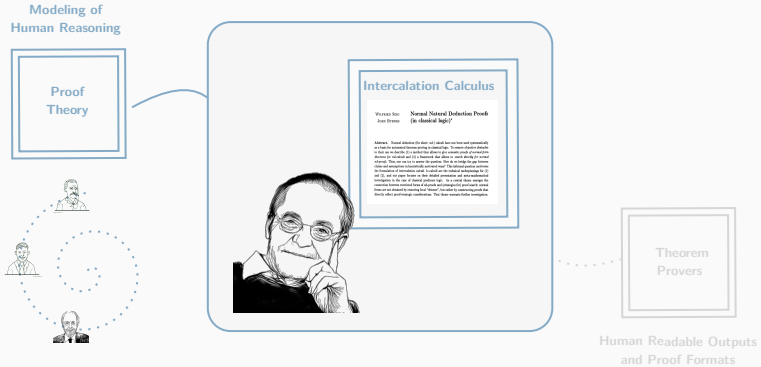
Proof Theory: Modeling of Human Reasoning

Natural Deduction enjoyed an awakening in 1965 with **Prawitz** and the (re)discovery of normalization, but kept suffering the anathema of being unsuitable for proof search.



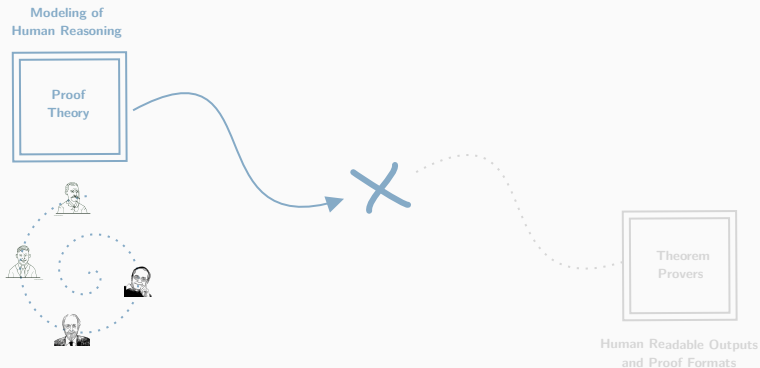
Proof Theory: Modeling of Human Reasoning

Sieg changed this state of affairs, in 1998, with his Intercalation Calculus. However, it has only been applied in an educational context, and seems to not have influenced theorem provers.



Proof Theory: Modeling of Human Reasoning

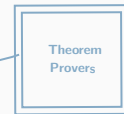
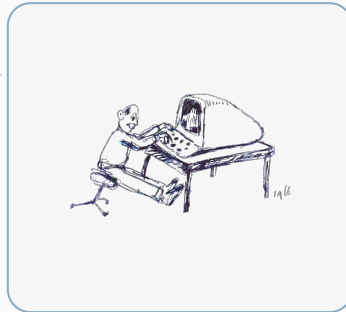
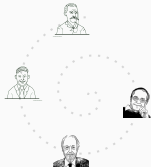
From the proof theory side, we aim to discover if we already have a **model of human reasoning** which may contribute to the goal of human-centered theorem proving.



Theorem Provers: Human Readable Outputs & Proof Formats

On the other hand, the interest and research in **automated theorem proving** arose, in the 1950s, in the domain of Computer Science and AI, but it only concerns machine, not human, reasoning.

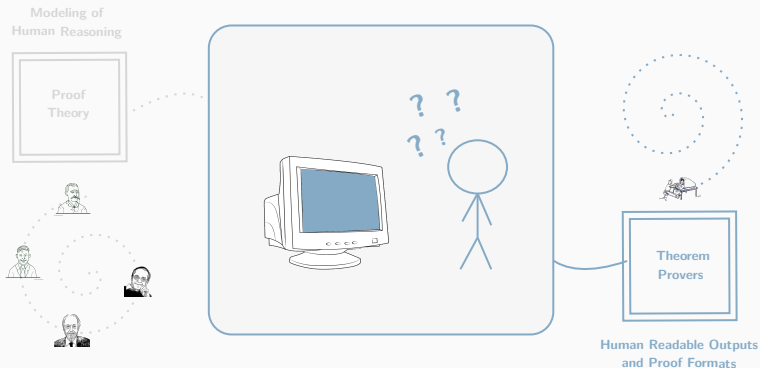
Modeling of
Human Reasoning



Human Readable Outputs
and Proof Formats

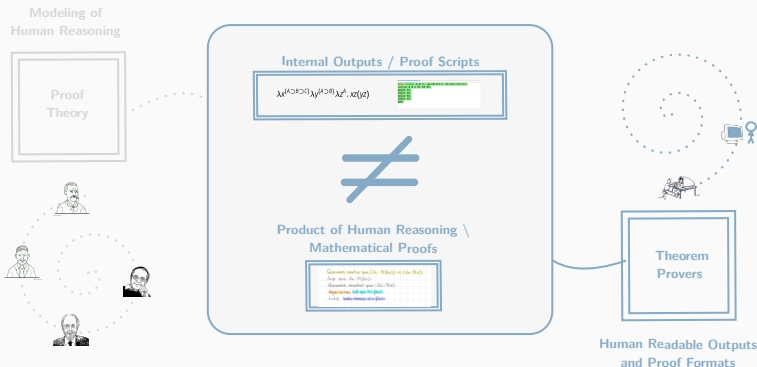
Theorem Provers: Human Readable Outputs & Proof Formats

The alternative architecture is the one of **interactive theorem proving**, in which a human user/prover/programmer is in dialogue with a proof checker.



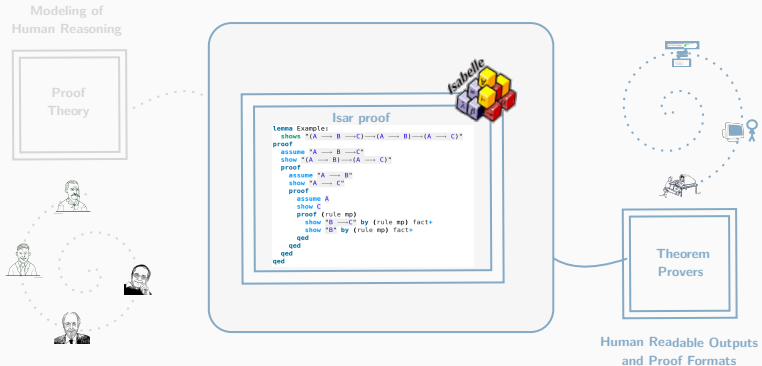
Theorem Provers: Human Readable Outputs & Proof Formats

However, neither the internal **outputs** nor the proof scripts that command the interactive development correspond to the product of human reasoning or resemble mathematical proofs.



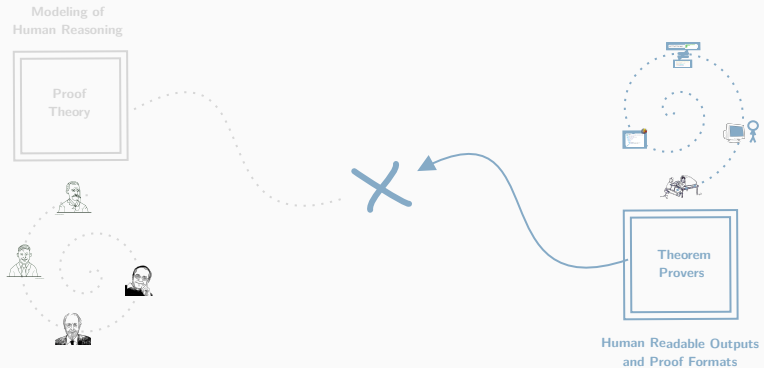
Theorem Provers: Human Readable Outputs & Proof Formats

As a possible answer to this question, in the context of *Isabelle* system, arose **Isar** - an environment for "declarative" natural deduction proofs that may be "executed" at the same time.



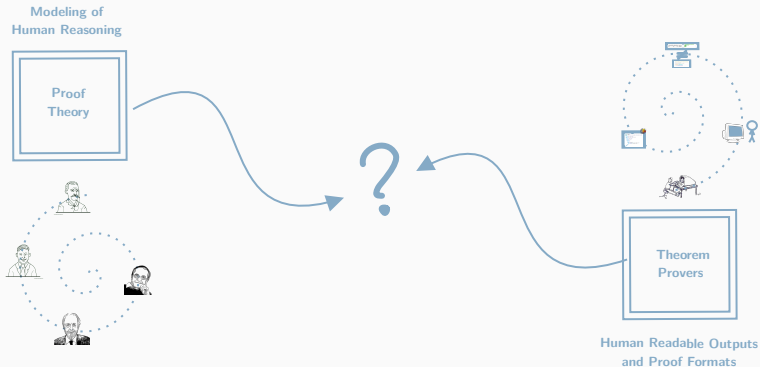
Theorem Provers: Human Readable Outputs & Proof Formats

From the theorem provers side, we will investigate if it is possible to reverse-engineer a **model of human reasoning** from *Isar*.



Proof Search in Natural Deduction: Open Question

Do these developments synthesize a
better model of human reasoning?



Aknowledgements

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